

Vitalii Nikitchyn

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SKILLS

Fields of interest: Software development (C++, Java, Kotlin, Android development), Data Analysis, Research

Computer Vision: OpenCV, Fourier transform, Convolution (img. processing operation), Canny edge detection, Pattern detection algorithms (applicable to a financial chart), Image restoration algorithms

C++ libraries/tools: C++ standard library (STL), Asio, Qt framework, CMake, POCO, UI tools (OpenGL, GLFW, ImGui), uSockets

Android libraries/tools: Gradle, Firebase, Retrofit, Apollo-graphql, Glide, Koin, Room, MVVM & MVP architecture, Unit testing

Databases: Neo4j (graph database) + Cypher QL, PostgreSQL + SQL

Platform independent skills: SOLID software design principles, Software design patterns, UML, JSON

Multithreading/Async: Concurrency & synchronization mechanisms (mutex, atomic variables, lock-based & lock-free data structures), compare-and-swap (CAS), event-loop, epoll, io-uring (linux kernel API for async I/O), multiplexing principle

Computer Networking: OSI network model, TCP/UDP/HTTP/WebSockets/ protocols, TLS, WebRTC

OS/IDE/Other tools: Linux OS (advanced daily user), IntelliJ IDE, Clion, Git, SVN, Gerrit, Jira/Confluence

Languages: English, French, Ukrainian, Russian

EDUCATION

Master's degree in Computer Science, September 2017 – December 2021.

University of Montreal, Montreal, Quebec, Canada.

- Thesis: [Domain-specific modeling on graph databases](#).
Related to: Neo-Modeling-Framework (NMF) – see detailed description in PROJECTS section

Bachelor's degree in Computer Science, September 2013 – June 2017.

Kyiv Polytechnic Institute, Kyiv, Ukraine.

WORK EXPERIENCE

Software Developer, Savoir-faire Linux, February 2024 – January 2025. Montreal, Canada.

- Support and Development of Jami application: <https://jami.net>
- Responsibilities: support & development of Jami android app and cross platform C++ module
- Android app Gitlab: <https://git.jami.net/savoirfairelinux/jami-client-android>
- C++ shared module Gitlab: <https://git.jami.net/savoirfairelinux/jami-daemon>

Android Developer, HEY CLAY, April – September 2021. Kyiv, Ukraine.

- Development of Pixio Android application: <https://pixio.site>
- Link on Google Play: <https://play.google.com/store/apps/details?id=site.pixio.Pixio>

PROJECTS

Neo Modeling Framework (NMF), Academic Project, September 2018 – September 2020

- A framework for domain-specific modeling and big-data manipulations on graph databases
- Delegates to the database handling of a large amount of data

- Ensures efficient data scalability and maintains consistency across distributed data graphs
- Generates a domain-specific API based on a domain metamodel
- Used technologies: Kotlin, Neo4j (graph database), Cypher query language, UML, Software architecture design, Distributed database design, Partial on-demand data loading, Bulk CRUD operations, Performance testing, Code generation.
- GitHub: <https://github.com/geodes-sms/NeoModelingFramework>

Video-based profiling, Academic project, September 2018 – October 2018

- A cross-platform desktop application: detects people in real-time video, acquires their features (age and gender), and stores that information in a database. In the event of a subsequent appearance by a particular person in front of the camera, the application retrieves the previously saved profile to recognize that person
- Used technologies: C++, Qt, OpenCV, Azure FaceAPI, GUI development, MySQL database
- GitHub: <https://github.com/vnikitch/Video-Based-Profiling>